

SOF IGKO — Unit 9 Deep-Dive

Maths Fun

Class III • Parent-readable notes

What this unit is about

Number patterns, the four basic operations, shapes, and simple logic puzzles — the kind of quick maths thinking the Olympiad rewards.

1. Numbers

- **Even numbers** end in 0, 2, 4, 6, 8 (2, 4, 6, 8, 10...).
- **Odd numbers** end in 1, 3, 5, 7, 9 (1, 3, 5, 7, 9...).
- **Comparing**: the number further along when counting is bigger ($12 > 7$).

2. The four operations

Operation	Sig n	Meaning	Example
Addition	+	Put together	$5 + 3 = 8$
Subtraction	-	Take away	$10 - 4 = 6$
Multiplication	×	Repeated addition	$3 \times 4 = 12$
Division	÷	Sharing equally	$12 \div 3 = 4$

3. Number patterns (find the rule)

- **Add the same number**: 5, 10, 15, 20, **25** (rule: +5).
- **Double each time**: 2, 4, 8, 16, **32** (rule: $\times 2$).
- **Square numbers**: 1, 4, 9, 16, **25** ($1 \times 1, 2 \times 2, 3 \times 3, \dots$).

4. Shapes

Shape	Sides	Corners
Triangle	3	3
Square	4 (all equal)	4
Rectangle	4	4
Circle	0 (round)	0

- A **cube** (like a dice) has **6 flat faces**.

5. Logic puzzles

- **Odd one out:** find the item that doesn't fit (2, 4, 6, **7** — 7 is odd).
 - **Analogy:** 2 is to 4 as 3 is to **6** (each is doubled).
 - **Sequence:** work out the rule, then find the next number.
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Common exam traps

- **0 is an even number.**
 - A **circle has no corners**; a **triangle has 3**.
 - Multiplication is **repeated addition** (not subtraction).
 - In “double each time” patterns, the jump keeps growing — check the rule before answering.
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Quick recall

1. Even numbers end in → **0, 2, 4, 6, 8**
2. $\times$ means → **multiply** (repeated addition); $\div$ means → **share equally**
3. Square → **4 equal sides**; triangle → **3 sides**
4. Circle corners → **0**; cube faces → **6**
5. 5, 10, 15, ... rule → **add 5**; 2, 4, 8, 16 rule → **double**